

EDA Report – Cryptocurrency Volatility Prediction

1. Introduction

This project analyzes cryptocurrency market data to understand price volatility patterns. Exploratory Data Analysis (EDA) is performed to study trends, correlations, and distributions in the dataset.

2. Dataset Description

The dataset contains historical daily data of multiple cryptocurrencies including:

- Open price
- High price
- Low price
- Close price
- Trading volume
- Market capitalization
- Date and timestamp
- Cryptocurrency name

3. Data Cleaning

Missing values were handled using forward filling.

Duplicate rows were removed.

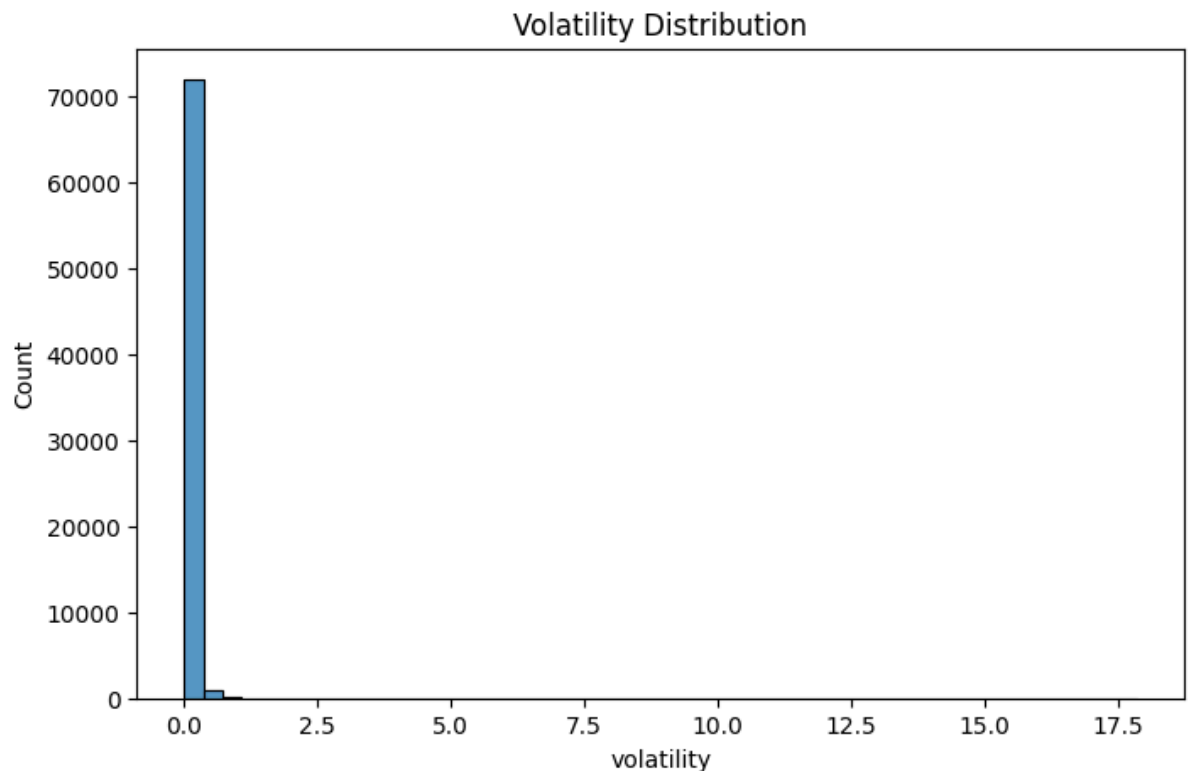
Irrelevant columns such as index, timestamp and crypto name were removed.

Infinite values were replaced with NaN and dropped.

4. Graph describe:-

1. Graph Title

Volatility Distribution



Description

This graph shows the distribution of cryptocurrency volatility values in the dataset. A histogram is used to visualize how frequently different volatility levels occur.

From the graph, it can be observed that most volatility values are concentrated near zero, indicating that the majority of cryptocurrency price movements have relatively small fluctuations. Only a small number of observations show very high volatility values.

The distribution is highly right-skewed, meaning that while most values are low, there are a few extreme values representing sudden and significant price changes in the cryptocurrency market.

2. Graph Title

Closing Price Trend of Cryptocurrencies



The closing price trend graph shows how cryptocurrency prices have changed over time. The x-axis represents the sequence of observations (time progression), while the y-axis represents the closing price of the cryptocurrency.

From the graph, it can be observed that cryptocurrency prices initially remained relatively low and stable. However, as time progressed, the prices started increasing significantly, showing strong upward movements during certain periods.

The graph also shows several fluctuations where prices rise rapidly and then fall, which reflects the highly volatile nature of cryptocurrency markets. Towards the later part of the graph, the price reaches very high levels compared to the earlier period, indicating a strong growth phase followed by corrections.

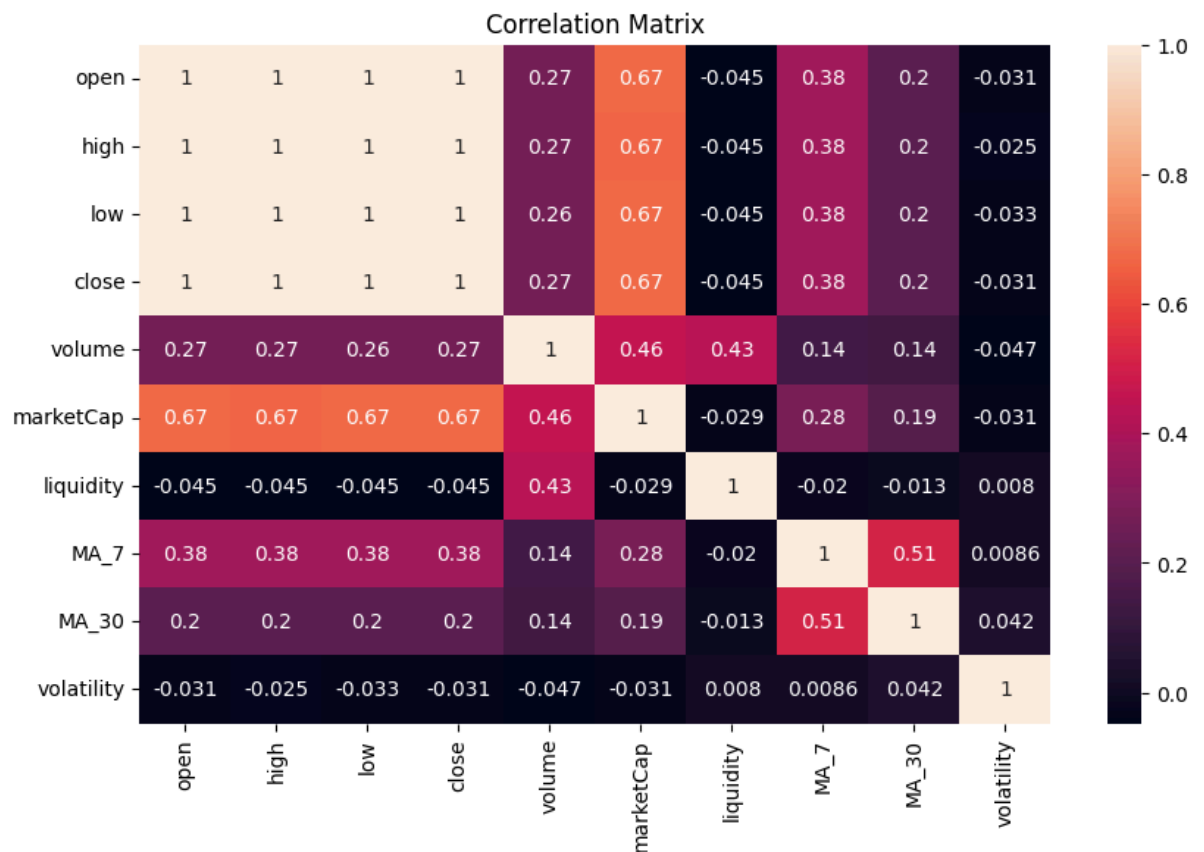
Overall, the graph demonstrates that cryptocurrency markets experience both rapid growth and sharp declines over time.

Key Insights

1. The closing price shows a long-term upward trend, indicating growth in the cryptocurrency market over time.
2. There are significant fluctuations, which highlight the volatile nature of cryptocurrency prices.
3. Periods of rapid price increase are followed by market corrections, where the price temporarily declines before stabilizing again.

3. Graph Title

Correlation Matrix of Cryptocurrency Features



Description

The correlation matrix heatmap shows the relationship between different numerical features in the cryptocurrency dataset. Each cell in the heatmap represents the correlation coefficient between two variables, with values ranging from -1 to 1.

A value close to 1 indicates a strong positive correlation, meaning that both variables increase together. A value close to -1 indicates a strong negative correlation, where one variable increases while the other decreases. Values close to 0 indicate little or no linear relationship.

From the heatmap, it is clear that the open, high, low, and close prices are highly correlated with each other, which is expected because they represent different price points of the same trading period. The market capitalization also shows a strong positive correlation with price-related features, indicating that as prices increase, market capitalization tends to increase as well.

The moving average features (MA_7 and MA_30) show moderate correlation with price-related variables, which suggests they capture overall price trends over time. On the other hand, volatility shows very weak correlation with most features, indicating that volatility may depend on sudden market movements rather than steady price trends.

Key Insights

1. Open, High, Low, and Close prices are strongly correlated, showing consistent price movement patterns.
2. Market capitalization has a strong positive relationship with price variables, meaning higher prices generally lead to higher market value.
3. Volatility has weak correlation with most features, suggesting that sudden market events influence volatility rather than gradual price trends.